

006037



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

MATHEMATICS

4008/2

PAPER 2

JUNE 2006 SESSION

2 hours 30 minutes

Additional materials:

- Answer paper
- Geometrical instruments
- Graph paper (3 sheets)
- Mathematical tables
- Plain paper (1 sheet)

ME 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **all** questions in Section A and any **three** questions from Section B.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

Electronic calculators must not be used.

All working must be clearly shown. It should be done on the same sheet as the rest of the answer. Omission of essential working will result in loss of marks.

If the degree of accuracy is not specified in the question and if the answer is not exact, the answer should be given to three significant figures. Answers in degrees should be given to one decimal place.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question. Mathematical tables may be used to evaluate explicit numerical expressions.

This question paper consists of 12 printed pages.

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Section A [64 marks]

Answer *all* the questions in this section.

- 1 (a) Express as a fraction in its simplest form $\frac{3}{2} + \frac{4x-1}{3}$. [2]
- (b) Given the formula $S = \frac{1}{2}n(r+l)$,
- (i) calculate the value of S when $n = 13$, $r = 7$ and $l = 11$,
- (ii) make r the subject of the formula. [4]
- (c) Calculate the principal that earns \$18 450 simple interest at 6% per annum in $1\frac{1}{2}$ years. [3]
- (d) Solve the equation $5(y+3) - 7 = 3(4-y)$. [3]

2

Harare

141	Kadoma		
275	134	Gweru	
439	x	164	Bulawayo
879	738	604	440
Victoria Falls			

The figure above shows a distance chart for five towns. The distances are in kilometres. From the chart the distance between Gweru and Victoria Falls is 604 kilometres.

- (a) (i) Write down the distance between Harare and Victoria Falls correct to the nearest 10 kilometres.
- (ii) Calculate x , the distance between Kadoma and Bulawayo. [3]
- (b) A motorist travels from Bulawayo to Victoria Falls at an average speed of 80 km/h.
- Calculate
- (i) the time the motorist takes to complete the journey,
- (ii) the fuel the motorist needs if the car uses 6,5 litres for every 100 kilometres travelled. [4]

- 4 Answer the whole of this question on a sheet of plain paper.

Use ruler and compasses only for all constructions and show clearly all the construction lines and arcs.

- (a) Construct on a single diagram

(i) a pentagon $PQRST$ in which $PQ = 6$ cm, $QR = 7$ cm, $RS = ST = 8$ cm, $PT = 4$ cm, $\hat{PQR} = 120^\circ$ and $\hat{QPT} = 135^\circ$,

(ii) the locus of points equidistant from ST and SR ,

(iii) a line through R parallel to PQ to cut ST at X .

[10]

- (b) State the reason why triangles PRQ and PXQ are equal in area.

[1]

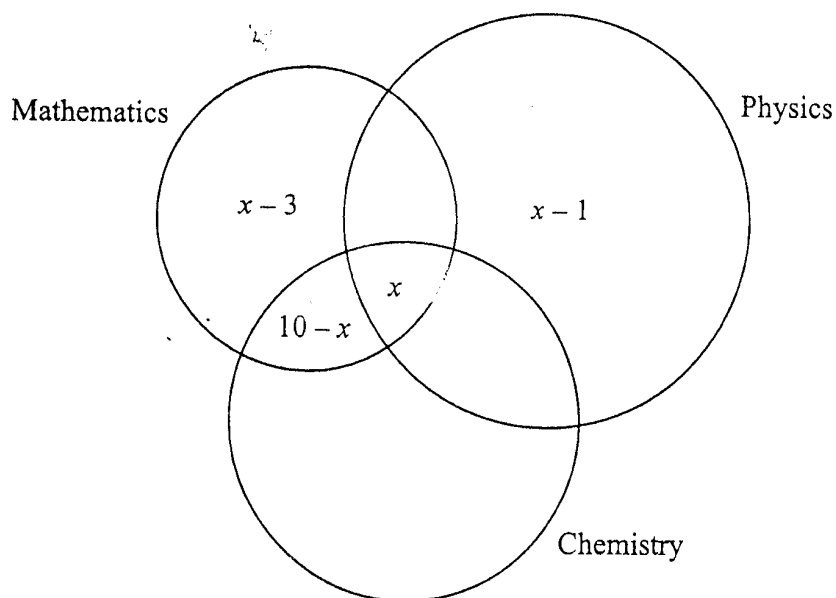
- 5 (a) In a group of 25 students, 15 study Mathematics, 12 Physics and 18 Chemistry. It is given that

8 study Mathematics and Physics,
5 study Physics and Chemistry,
10 study Chemistry and Mathematics and
 x study all the three subjects.

Each student studies at least one of these subjects.

- (i) Copy and complete the Venn diagram to show the number of elements in each subset in terms of x .

910
65
35
60



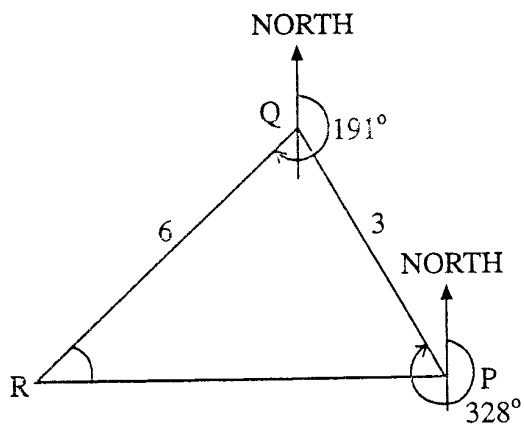
- (ii) Form an equation in x and solve it. Hence or otherwise find the number of students who study Mathematics and Chemistry **only**. [6]
- (b) Two fair-sided coins are tossed at the same time and the results noted.
- (i) Using the symbols H for Heads and T for Tails, list all the four possible outcomes.
- (ii) Write down the probability of getting two Heads or two Tails. [3]
- (c) Solve the equation

$$2^{2n-1} = 64.$$

[2]

Section B [36 marks]

Answer *three* questions in this section.



The diagram shows three villages P, Q and R. The bearing of Q from P is 328° and the bearing of R from Q is 191° . $PQ = 3$ km and $QR = 6$ km.

(a) Calculate

(i) $\hat{PQR}$,

(ii) the distance Q is north of P,

(iii) PR.

[8]

(b) Find the bearing of R from P, giving your answer correct to the nearest degree.

[4]

8 Answer the whole of this question on a sheet of graph paper.

A particle moves along a straight line so that after t seconds its velocity, v m/s, is given by the formula

$$v = 5 + 7t - 2t^2.$$

Corresponding values of t and v are given in the table below.

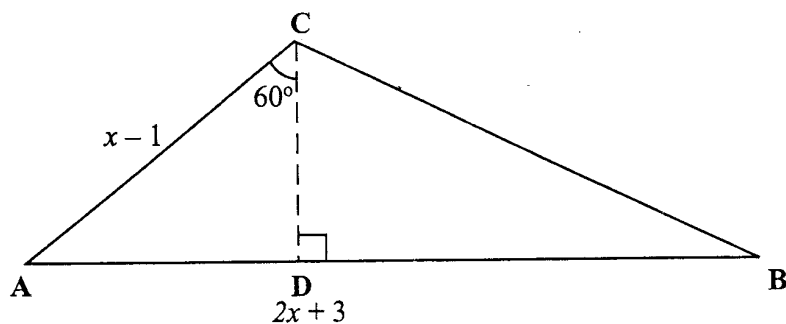
t (s)	0	1	2	3	4	5
v (m/s)	5	10	11	8	p	-10

(a) Calculate the value of p . [1]

(b) Taking 2 cm to represent 1 second on the horizontal axis and 2 cm to represent 4 m/s on the vertical axis, draw the graph of $v = 5 + 7t - 2t^2$ for $0 \leq t \leq 5$. [4]

(c) Use the graph to estimate,

- (i) the maximum velocity of the particle,
- (ii) the value of t when the particle is stationary,
- (iii) the acceleration of the particle when $t = 5$,
- (iv) the distance the particle moves between $t = 2$ and $t = 4$. [7]



In the diagram $AC = (x - 1)$ cm, $AB = (2x + 3)$ cm, $\hat{BDC} = 90^\circ$ and $\hat{ACD} = 60^\circ$.

- (a) Find, in terms of x , the length of CD. [2]
- (b) Write down an expression for the area of triangle ABC and show that it reduces to $\frac{1}{4}(2x^2 + x - 3)$. [2]
- (c) (i) Given that the area of triangle ABC is 9 cm^2 , form an equation in x and show that it reduces to $2x^2 + x - 39 = 0$.
- (ii) Solve the equation $2x^2 + x - 39 = 0$ giving your answers correct to two decimal places.

Hence write down the length of AC. [8]

10 Answer the whole of this question on a sheet of graph paper.

Using a scale of 2 cm to represent 2 units on each axis draw the x and y axes for $-4 \leq x \leq 16$ and $-12 \leq y \leq 8$.

- (a) Triangle ABC has vertices at A(0; 3), B(2; 1) and C(4; 5). Draw and label triangle ABC. [1]
- (b) Triangle ABC is mapped onto triangle $A_1B_1C_1$ by a reflection in the line $y = 0$.
- (i) Draw and label triangle $A_1B_1C_1$.
- (ii) Write down the matrix that represents this reflection. [4]
- (c) Triangle $A_2B_2C_2$ has vertices at $A_2(6; 3)$, $B_2(4; 1)$ and $C_2(14; 5)$.
- (i) Draw and label triangle $A_2B_2C_2$.
- (ii) Describe fully the single transformation which maps triangle ABC onto triangle $A_2B_2C_2$. [4]
- (d) Triangle ABC is mapped onto triangle $A_3B_3C_3$ by an enlargement of scale factor -2 and centre (5; 0). Draw and label triangle $A_3B_3C_3$. [2]
- (e) Find the ratio $\frac{\text{Area of triangle ABC}}{\text{Area of triangle } A_3B_3C_3}$. [1]
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- 11 (a) Students in a mathematics class each contributed some money towards charity. The amounts were grouped as shown in the table below.

Amount (\$)	$15 < x \leq 20$	$20 < x \leq 23$	$23 < x \leq 27$	$27 < x \leq 34$	$34 < x \leq 40$
Frequency	12	11	8	10	4

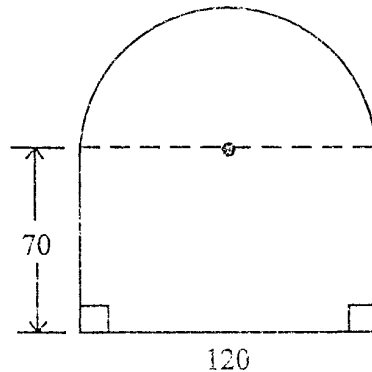
(i) Find the number of students in the class.

(ii) Write down the modal group.

(iii) Calculate the mean amount contributed.

[5]

- (b) In this question take π to be 3.142.



The diagram above shows a scale drawing of an advertising billboard. It is made up of a rectangle of length 120 cm and width 70 cm and a semicircular top.

Calculate

(i) the area of the scale drawing,

(ii) the area of the actual billboard given that the billboard is 3 metres long,

(iii) the cost of painting the actual billboard at \$4 000 per square metre. [7]

12 Answer the whole of this question on a sheet of graph paper.

A factory produces two types of desks, type A and type B. It produces x desks of type A and y desks of type B per year.

- (a) Given that the factory produces at most 5 000 desks annually, write down an inequality in x and y which satisfies this condition. [1]
- (b) The number of desks produced annually is such that the number of type B desks is not less than half the number of type A desks and is also not more than the number of type A desks. Write down **two** inequalities in x and y which satisfy these two conditions. [2]
- (c) Given also that it costs \$4 000 to make a type A desk and \$9 200 to make a type B desk and that the factory has at least \$18 400 000 for its annual production, write down an inequality in x and y which satisfies this condition and show that it reduces to $10x + 23y \geq 46\,000$. [2]
- (d) Taking 2 cm to represent 500 desks on each axis, draw the x and y axes for $0 \leq x \leq 4\,000$ and $0 \leq y \leq 3\,500$. Draw the graphs of these inequalities and show, by shading the **UNWANTED REGIONS**, the region in which x and y must lie. [5]
- (e) If the profit on a type A desk is \$1 200 and on a type B desk is \$1 000, find the number of each type of desk the factory should produce in order to obtain the greatest profit. [2]
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